MECH191—Metallurgy

Week 10

Casting Processes & Defect Recognition

October 26, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

Solidification & Sand Casting

How metal solidifies — shrinkage, gas, segregation, risers, chills, dendrites — and the sand casting process from pattern to shakeout

02

Die Casting & Investment Casting

High-pressure die casting, hot vs. cold chamber, lost-wax investment casting, and other casting processes

03

Casting Defects & Process Selection

Recognizing shrinkage porosity, gas porosity, hot tears, cold shuts, misruns, and inclusions — and how to choose the right process

Metallography Lab — Module 9

EXAMET-5-R-600-HD · 50–500×

Module 9 · Week 10

Wrought Aluminum Alloys

What to do this week:

Compare constituent particle size/density between 6061-T6 and 2024-T3 at 500×. This week's lab no longer uses a cast specimen — casting fundamentals remain in today's lecture.

Steps:

1 Observe — complete feature table
2 Sketch — label magnification & specimen
3 Answer — 4 short questions

Specimens to Check Out

· Specimens G, J — 6061-T6 & 2024-T3 aluminum

How to check out the kit:

1 See instructor after class today
2 Sign out the specimen tray
3 Use the EXAMET-5 at your convenience
4 Return kit before leaving class
5 Submit completed module with your portfolio

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 ← here Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 Sep 21 Stainless & Cast Irons	Wk 8 Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

Week 10 — At a Glance

Topics

- Solidification mechanics — shrinkage, gas porosity, segregation, dendrite arm spacing, and how riser and chill design controls them
- Sand casting — pattern, mold, gating system, cores, green sand vs. chemically bonded sand, advantages and limitations
- Die casting — hot chamber vs. cold chamber, high-pressure injection, surface finish and tolerance advantages, alloy limitations
- Investment (lost-wax) casting — ceramic shell process, best surface finish and accuracy of any casting process, applications
- Seven common casting defects — shrinkage porosity, gas porosity, hot tears, cold shuts, misruns, inclusions, and warping
- Process selection — alloy, part size, production volume, required tolerance, and cost as the five deciding factors

Resources

Reference Material

Kalpakjian Ch. 10 & 11
Fundamentals & Processes of Metal Casting

Visualizers

Solidification & Dendrite Growth
Sand Casting — Process & Defects
Die Casting — NADCA Process
Investment Casting — ICI Process

Reference

Casting Defects — TWI Defect Library

Next: Week 11 — Metal Forming

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What happens during solidification and why does it matter for casting quality?
2. What are the main casting processes and what are the trade-offs between them?
3. What defects are most common in castings and how is each one recognized?
4. How do you choose the right casting process for a given application?

Solidification & Shrinkage — What Happens When Metal Freezes

Solidification Zones

Chill Zone

Fine equiaxed grains nucleate at mold wall — fastest cooling

Columnar Zone

Dendrites grow inward toward thermal center — directional heat flow

Equiaxed Zone

Last region to solidify — equiaxed grains; site of shrinkage & segregation

Key Mechanisms

Shrinkage

Metal contracts 2–8% vol. during solidification. Risers (feeders) supply liquid metal to compensate — without a riser, internal voids or surface sink marks form in the last section to solidify.

Gas Porosity

Dissolved gases (especially H_2) become insoluble as metal solidifies and are expelled. If they cannot escape, smooth spherical voids form — distributed through the casting or clustered near the surface.

Segregation

Alloying elements partition between solid and liquid — low-melting elements concentrate in the last liquid, creating compositional gradients. Homogenization heat treatment reduces microsegregation.

Dendrites & DAS

Solidification begins as tree-like dendrites growing from the mold wall. Dendrite arm spacing (DAS) decreases with faster cooling — finer DAS means better mechanical properties.

Sand Casting — The Most Flexible Casting Process

1 Pattern Making

Replica of part in wood, plastic, or metal — includes shrinkage, machining, and draft allowances

2 Mold Prep

Sand (green or chemically bonded) packed around pattern in a flask; pattern withdrawn leaving the cavity

3 Core Making

Separate sand shapes inserted to create internal cavities or undercuts the pattern cannot form

4 Mold Assembly

Cope (top) and drag (bottom) halves assembled; gating system — sprue, runner, gates — channels metal in

5 Pouring

Molten metal poured into pouring cup; controlled pour rate prevents turbulence and oxide entrapment

6 Cooling

Metal solidifies; risers feed shrinking sections; chills can accelerate cooling in thick sections

7 Shakeout

Sand mold is broken away; casting removed; sand reclaimed for reuse

8 Cleaning & Finish

Gates and risers removed; surface cleaned by shot blasting; dimensional and visual inspection

Advantages

Low tooling cost · Any metal and almost any size · Complex internal geometry with cores · Short lead time at low volume

Limitations

Rough surface finish (Ra 6–25 µm) · Loose dimensional tolerances · Labor intensive · Each mold destroyed after shakeout

Die Casting — High Pressure, High Volume, Tight Tolerances

Hot Chamber Die Casting

- Injection mechanism submerged in molten metal bath
- Fast cycle times — mechanism always ready with liquid metal
- Limited to low-melting-point alloys: zinc, tin, lead
- Cannot use for aluminum — molten Al attacks the steel mechanism

Cold Chamber Die Casting

- Metal ladled into separate shot chamber before each injection
- Slower cycle than hot chamber but handles higher-melting alloys
- Used for aluminum, magnesium, copper — the important structural alloys
- Injection pressure: 7–700 MPa; high velocity fills complex die cavities

Advantages

Excellent surface finish (Ra 0.8–3.2 µm) · Tight tolerances · Fast cycle times · Thin walls possible

Limitations

High tooling cost · Limited to lower-melting alloys · No sand cores — limited internal geometry

Die Casting Porosity

High velocity traps air — gas porosity is inherent in die castings. Limits pressure-tight and heat-treat applications. Vacuum die casting significantly reduces this.

Investment Casting — Lost Wax, Best Accuracy

1 **Wax Pattern**
Inject wax into a metal die — produces an exact wax replica of the finished part

2 **Tree Assembly**
Multiple wax patterns attached to a central wax sprue forming a 'tree'

3 **Shell Building**
Tree dipped in ceramic slurry, stuccoed with refractory sand — repeated 5–15 times

4 **Dewaxing**
Steam autoclave melts and removes the wax — leaving a hollow ceramic shell

5 **Shell Firing**
Kiln burns residual wax, sinters ceramic — shell is now rigid and ready for pouring

6 **Pouring**
Molten metal poured into pre-heated shell — ceramic conforms exactly to wax surface

7 **Shell Removal**
Ceramic shell broken away; parts cut from tree, finished, inspected

Why Investment Casting?

- Best surface finish of any casting process — Ra 1.6–3.2 μm
- Tightest tolerances — ±0.1–0.25mm on small features
- Almost any alloy, including superalloys and reactive metals
- Complex geometry, thin walls, undercuts — often without cores
- Applications: turbine blades, medical implants, firearm parts, jewelry

Other Casting Processes

Permanent Mold: Gravity pour into reusable metal mold — better finish than sand, limited alloys

Centrifugal: Rotating mold distributes metal outward — used for pipes, rings, tubes; dense outer surface

Continuous Casting: Molten steel/aluminum solidified continuously into slabs and billets — dominant process for structural steel

Common Casting Defects — Recognition and Detection

Defect	Cause	Appearance	Likely Location	Detection
Shrinkage Porosity	Insufficient feed metal; poor riser design or placement	Irregular, rough-surfaced spongy voids	Thick sections; last to solidify	RT, UT
Gas Porosity	Dissolved hydrogen expelled during solidification; moisture in mold	Smooth, spherical voids — distributed or clustered	Throughout casting or near surface	RT, UT
Hot Tears	Mold restrains contracting casting during solidification	Irregular branching cracks with oxidized surfaces	Section changes, sharp corners	VT, PT, MT
Cold Shuts	Two metal streams meet but too cold to fuse; low pouring temp	Linear seam with smooth oxidized surface	Mid-section; opposite ingate	VT, PT
Misrun	Metal solidifies before filling mold; thin section or low temp	Incomplete casting — smooth rounded edges at termination	Thin sections; far from ingate	VT
Inclusions	Sand, slag, or oxide entrained during pouring	Irregular hard spots; dark irregular RT indications	Near gates; upper mold surfaces	RT, UT
Warping	Non-uniform cooling; residual thermal stress	Dimensional distortion — not visible as a crack or void	Long flat sections	Dimensional

Casting Process Selection — Five Factors, One Right Answer

Factor	Sand Casting	Die Casting	Investment Casting
Alloy	Any metal — steel, cast iron, aluminum, copper, superalloys	Lower-melting only — Al, Zn, Mg, Cu; NOT steel or cast iron	Almost any — including superalloys and reactive metals
Part Size	Virtually unlimited — from grams to hundreds of tonnes	Limited by machine capacity — typically under 25 kg	Typically under 50 kg; best for small to medium precision parts
Production Volume	Low to medium — tooling cost low; each mold is single-use	High volume only — die cost (tens of $K) must be amortized	Low to medium — ceramic shell is single-use but patterns are reusable
Tolerance & Finish	Rough — Ra 6–25 µm; tolerances ±1–3 mm on large parts	Good — Ra 0.8–3.2 µm; tolerances ±0.1–0.3 mm	Best — Ra 1.6–3.2 µm; tolerances ±0.1–0.25 mm on small features
Internal Geometry	Excellent — sand cores create complex internal passages	Limited — no sand cores; some complexity with die slides	Good — many undercuts possible without cores; shell broken away

Technician's Perspective

Reading the defect — what it tells you about how the casting was made.

Red Flag: Die casting ordered for a 304 stainless steel bracket

A job order specifies die casting for a 304 stainless steel structural bracket. Do not proceed — die casting cannot be used for stainless steel. The steel die cannot withstand the pouring temperature of stainless (above 1400°C), and the metal would freeze before filling the die. Die casting is limited to aluminum, zinc, magnesium, and copper alloys. Correct the process specification: investment casting or sand casting are the appropriate processes for this part.

Caution: Specifying RT on a die casting to find internal defects

A die cast aluminum housing is being evaluated for a pressure-tight application. RT reveals no defects — do not accept this as proof of soundness. Die castings inherently contain gas porosity from air trapped during high-velocity injection; this porosity is often fine, distributed, and below RT detection sensitivity. UT is more sensitive to this type of defect. Additionally, confirm the application: die cast parts with this porosity may blister if subsequently heat treated. Consult the applicable specification before accepting.

Good Practice: Identifying hot tears vs. cold cracks by oxidation

A cracking indication is found on a sand casting at a thick-to-thin section change. To distinguish a hot tear from a cold crack: hot tears form during solidification when the metal is still partly liquid — the fracture surfaces are oxidized (dark, discolored) because they were exposed to atmosphere at high temperature. Cold cracks form after solidification — the surfaces are clean and metallic. An oxidized crack surface at a stress-concentration point is a hot tear and indicates a casting design or mold restraint issue that must be corrected before re-casting.

Core Question Answers

What happens during solidification and why does it matter for casting quality?

Solidification starts at the mold wall (chill zone), progresses inward as columnar dendrites, and finishes in the equiaxed center — the last zone to solidify is where shrinkage voids, gas porosity, and segregation concentrate. Riser design, chill placement, and pouring temperature control where and when solidification occurs, and therefore where defects form.

What are the main casting processes and what are the trade-offs between them?

Sand casting: any alloy and size, low tooling cost, rough surface. Die casting: high pressure, excellent finish, limited to lower-melting alloys, high tooling cost, gas porosity inherent. Investment casting: ceramic shell over wax pattern, best surface finish and accuracy, any alloy, expensive per part, limited to smaller components.

What defects are most common in castings and how is each one recognized?

Shrinkage porosity: irregular rough voids in thick sections (RT, UT). Gas porosity: smooth spherical voids (RT, UT). Hot tears: branching oxidized cracks at section changes (VT, PT, MT). Cold shuts: smooth linear seam where metal streams didn't fuse (VT, PT). Misrun: incomplete casting with rounded smooth edge (VT). Inclusions: hard spots with irregular RT indications (RT, UT).

How do you choose the right casting process for a given application?

Five factors together: alloy (die casting cannot handle steel; investment casting handles superalloys), part size (sand for large, die for small), production volume (die only justified at high volume), required surface finish and tolerance (investment for precision), and internal geometry complexity (sand with cores for complex passages).

Key Takeaways — Week 10: Casting Processes & Defect Recognition

1. Solidification is not instantaneous or uniform — shrinkage, gas evolution, segregation, and dendrite growth occur simultaneously, and controlling them through riser design, chilling, and pouring practice is what separates a sound casting from a defective one.

2. Sand casting, die casting, and investment casting each occupy a distinct niche defined by alloy, size, volume, tolerance, and cost — selecting the wrong process for an application leads to either excessive cost or unacceptable quality, and a technician who understands why a process was chosen can anticipate where defects are most likely.

3. Casting defects are predictable from the process — shrinkage porosity in thick sections, gas porosity in die castings, hot tears at section changes, cold shuts at ingate-remote locations. A technician who understands how a casting was made focuses inspection effort accordingly and interprets indications in the context of the manufacturing process.

Resources & What's Next

Week 10 Resources

Kalpakjian Ch. 10 & 11
Fundamentals of Metal Casting & Metal Casting Processes — primary reading

Casting Defects — TWI
Defect library with photographs and NDT detection guidance for each defect type

Solidification Visualizer
Dendrite growth, riser design, and solidification sequence — interactive

Sand Casting Visualizer
Process animation, design guidelines, and common defects — interactive

Die & Investment Casting
NADCA and ICI process animations with design and defect coverage — interactive

Week 11 — Metal Forming

November 2, 2026 · Unit 4: Manufacturing continues

- Hot working vs. cold working — the recrystallization temperature and why it defines the boundary between the two
- Work hardening — how dislocations accumulate during cold working and how annealing restores ductility
- Rolling, forging, extrusion, and drawing — what each process produces and where it is used
- Grain flow in forgings — why a forged part outperforms a casting or a machined bar in critical applications
- Forming defects — laps, seams, bursts, pipe defects, and delamination and how each is detected